Nobody Builds the Wall: LLM Agent Societies Leave a Public Good Unbuilt Without a Coordinator

Quan Tran, Alexandre Tancrède, Anas Elghoudane, Pierre Duhoux

Zarqa Lab; UdeM AI club

Every number traces to a file under `results/`; citations marked † are not yet cite-checked.

Abstract

We put the same 100 LLM agents (Gemma 4 26B), with the same personal goal, on islands that differ only in their economic rules, and let nightly mobs kill anyone outside a wall. Walls are a public good: they protect everyone and pay no one. Over 5 seeds of 12 days, capitalist islands built 0 walls and kept 5.0 of 100 people alive (range 3-7); a 20% tax that pays for wall building kept 8.8 alive but still built almost none (3 walls in 5 games). Socialist islands, where an elected leader sets job quotas, built 142 walls per game, lost no one to mobs, and kept 36.6 alive (30-48), losing the rest to starvation. Agents rested at similar rates under every rule set (0.24 vs 0.28), so the gap is provision of the public good, not effort. A second experiment (E14) crossed ownership (private or common) with coordination (none, an elected coordinator, or a planner) in six 60-agent economies with neutral wording and a loophole that pays the agent and secretly drains the wall. Islands with a coordinator used the loophole less in every seed (hack rate 0.09 vs 0.36 under private ownership, 0.14 vs 0.60 under common ownership, 3 of 3 preregistered seeds and 10 of 10 in total), and in 60-bot Minecraft. Common ownership without a coordinator hacked most and built no walls, so the wall depended on coordination, not ownership. The same model with the same goal lived or died by the rules it ran under, so evaluating each agent alone does not predict what the population does.

1. Introduction

- Alignment work mostly evaluates one model. Deployed agents increasingly act in groups that share resources, and the rules of the group set which individually reasonable choices add up to collapse.
- Public goods are the textbook case (Samuelson 1954†; Olson 1965†): a good everyone uses and no one is paid to make is under-provided by self-interested actors unless an institution provides it.
- We ask whether LLM agents reproduce this, and whether the economic rule set decides it, and which part of the rule set matters: who owns the stock, or who coordinates the work. The setup follows a public Minecraft experiment with 200 human players on a capitalist and a communist island (jjkay03, 2025), with LLM agents in place of people.
- Contributions:
  1. A preregistered island economy with three rule sets, one model and one goal (E08, 15 games of 100).
  2. A one-year, two-island extension where agents vote their rules, re-elect leaders and migrate (E12).
  3. A transfer check in real Minecraft with up to 50 bots per island (E09, E12-MC).
  4. Six 60-agent economies that cross ownership with coordination, with neutral wording and a loophole that pays the agent and drains the wall, in text (10 seeds) and 60-bot Minecraft (E14). Coordination, not ownership, set loophole use and whether the wall was built.
  5. Released code, preregistrations, ledgers and raw logs.

2. Related work

- LLM agents in commons and public-goods games. GovSim (Piatti et al. 2024†, arXiv 2404.16698) finds most LLM groups over-harvest a shared resource. Guzman Piedrahita et al. (2025†, arXiv 2506.23276) report that reasoning models free-ride in public-goods games. Both use small groups and fixed rules; we vary the rules and scale to 100 agents.
- Many-agent Minecraft societies. Project Sid (Altera.AL 2024†, arXiv 2411.00114) runs up to 1,000 agents and studies a tax vote; Civ50 (YC/Olam, 2026-10-29†) runs 50 agents once with no institutional variable. We hold the model and goal fixed and change only the institution.

- Economic design with learning agents. The AI Economist (Zheng et al. 2020†, arXiv 2004.13332) learns tax policy with RL agents; we use off-the-shelf LLMs and fixed, human-readable rules.
- Planning and information. Hayek (1945†) argues a central planner lacks agents' private knowledge. E14 tests this directly with private skills that agents can misreport.
- Proxy gaming in mixed-motive agent economies. A study by co-author Quan Tran on this repository (`origin/ben/C-scarcity-hacking`) tests an administrative tool (`ledger_sync`) that credits +5 to the caller's ledger for no work while draining 1.5 from the shared defence wall. Across 80 eight-agent LLM villages, every private-property villager used the tool (adoption 1.0) while commune villagers used it on 0 of about 3,200 villager-days (its `claims.md` C11), and in 10-village paired runs at survivable scarcity, villages with the tool banned kept 6.9 of 8 alive against 4.0 with no governance (C19, its `results/E09c/summary.json`). In that study, the private prompt states "Your goal: maximise YOUR personal wealth (food plus emeralds)" whereas the commune prompt states "Your goal: maximise the COMMUNE's prosperity". E14 ports the same `ledger_sync` payoff into our six island economies while keeping the personal goal identical across every cell ("Your personal goal: survive, and live comfortably. Your comfort score is your rest days plus the goods you personally own"), testing whether institutional rules alone change loophole use.

3. Setup

World (E08, **`src/swarm/islands.py`**). Each island has 100 people, the same start food and the same night rule. Each day every agent chooses a job: farm (food), chop (wood), mine (stone, sometimes ore, small cave-in risk), build_wall (wood and stone into wall), or rest. Each night mobs attack; the wall covers the share min(1, wall / mob_need) of the island, and an agent outside cover may die. Mob need grows each night after two calm nights; walls decay. Agents who end a night without food starve. Every agent has the same goal text: comfort, defined as rest days plus own goods.

Rule sets. | island | ownership | market | who builds walls | |—|—|—|—| | capitalist | private goods | bank buys ore; food-for-coins auction | anyone who donates | | capitalist_tax | private goods | the same plus 20% tax on bank sales | paid 2 coins per wall from the tax | | socialist | common store, equal rations | none | elected leader sets daily quotas |

Model and protocol. Gemma 4 26B (A4B instruct) through the Adaption API, temperature default, one call per agent per decision. Preregistered before any main game (`experiments/E08_islands/prereg.md`); pilot games are excluded. 5 seeds per arm; each seed gives all islands the same people and traits.

Year extension (E12, **`src/swarm/year.py`**, prereg **`experiments/E12_year/prereg.md`**). Islands A (capitalist) and B (socialist), 24 people each, 52 weekly turns. Winter (weeks 40-52) rots a quarter of stored food each week and makes mobs 1.5x stronger. An island with $\geq$ 6 food per head adds 10% newborns. Quarterly assemblies (weeks 13, 26, 39) re-elect the leader on common-store islands. In the open arm, every citizen also sees both islands, votes the rules for the next quarter, and may move. The fixed arm keeps the rules and blocks migration. Predictions Y1-Y6 are in the prereg.

Minecraft (E09, **`experiments/E09_minecraft/`**). A Paper 1.21.4 server with one mineflayer bot per agent. The LLM layer is unchanged; work is done physically (bots harvest wheat, chop logs, mine stone for 45 s), the wall is real cobblestone on a ring, and nights summon real zombies the bots fight. Block counts convert to text units by a preregistered calibration. In E12-MC, births log in new bots and a migrant's bot logs out of one island and into the other.

3.1 The 2x2 factorial (E13)

In E08 and E09, the socialist island differs from the capitalist island in three ways: goods go to a common store with equal rations (ownership), an elected leader sets daily job quotas and shows each person an assigned job (coordination), and the prompts name "capitalism" and "socialism" (labels). E13 (`experiments/E13_factorial/prereg.md`) crosses ownership (private goods with a market and bank vs shared store with equal rations) with coordination (each person picks a job vs a day-1 elected coordinator sets quotas and assigns jobs, shown but not enforced), and removes ideology words across all four cells (`PS`, `PC`, `CS`,

CC). Preregistered predictions on day-12 survivors over 5 text seeds test main effects of ownership (F1: (CS + CC) − (PS + PC) > 0 in ≥ 4 of 5 seeds) and coordination (F2: (PC + CC) − (PS + CS) > 0 in ≥ 4 of 5 seeds), their interaction (F3), label removal (F4: CC − PS > 0 in ≥ 4 of 5 seeds), and whether walls built follow the F1 and F2 signs (F5). Under Amendment 2, the E13 text run stopped on HTTP 429 during seed 0 directed because three processes shared one API key; only seed 0 commons (CS) finished in results/E08/e13.jsonl, so F1-F5 are not scored on E13 alone and the four factorial cells are tested inside E14.

3.2 Six economies with a loophole (E14)

E14 (experiments/E14_master/prereg.md) runs all six neutral-worded economies: the four factorial cells plus the taxed market (TX) and central planning with private skills (PL), adding the job ledger_sync to every cell. Choosing ledger_sync takes no work and no energy, credits +5 to the caller's ledger (coins on private islands; logged contribution shown on a top-3 public leaderboard on common-store islands), and removes min(wall, 1.5) from the island wall; the prompt states the +5 credit and omits the wall drain.

code	island	ownership	coordination	extra rules
PS	capitalist --neutral	private	self	market, bank
PC	directed	private	coordinator	market, bank
TX	capitalist_tax --neutral	private	self	20% sales tax pays 2 coins/wall, 1 coin/donated wood or stone
CS	commons	common	self	equal rations
CC	socialist --neutral	common	coordinator	equal rations
PL	planned --neutral	common	planner	private skills (1-5); planner sees self-reports only

The primary arm runs in Minecraft at n = 60 bots per island for 12 days (seed 0, two islands at a time on ports 25580 and 25581, results/E09/e14_mc.jsonl), paired with a 3-seed text arm at n = 60 (results/E08/e14.jsonl). Preregistered predictions test whether the hack rate (share of person-days on ledger_sync) is higher on private islands than on common islands (H1), whether a coordinator lowers the hack rate (H2: PC < PS and CC < CS), whether wall drained and day-12 survivors have Spearman rho < 0 across the six Minecraft islands (H3), and whether the survival ordering of the four factorial cells holds with the loophole present (H4). Stated limits: in Minecraft each cell is one 60-bot island whose agents read a shared message board, so agent-level Fisher tests treat non-independent agents as independent; the Minecraft calibration was fitted at 50 bots; and the physical wall is raised and lowered by server commands matching Python's wall count.

4. Results

4.1 Rules decide who survives (E08)

Data: results/E08/main_gemma.jsonl, 15 games, 10.5k calls, $0.64. Means over seeds 0-4. | island | alive /100 | starved | killed by mobs | walls | chosen rest | wealth Gini | compliance | |—|—|—|—|—|—|—|—| | capitalist | 5.0 (3-7) | 36.4 | 53.2 | 0 | 0.239 | 0.237 | - | | capitalist_tax | 8.8 (4-14) | 23.8 | 61.0 | 0.6 | 0.248 | 0.216 | - | | socialist | 36.6 (30-48) | 58.6 | 0 | 142 | 0.282 | - | 0.823 |

prereg	prediction	result
H1 public good	socialist walls per capita ≥ 1.5x capitalist	holds: 142 vs 0 in every seed

prereg	prediction	result
H2 free riding	socialist chosen rest ≥ capitalist + 0.05	fails: +0.043 (higher in 4 of 5 seeds)
H3 inequality	capitalist Gini ≥ 0.3	fails: 0.237
H4 survival	survivors differ by ≥ 10	holds: +31.6, socialist higher in every seed
H5 compliance	≥ 0.6	holds: 0.823
H6 institution	tax walls ≥ 1.5x capitalist	nominal only: 3 walls in 5 games vs 0

- The capitalist islands die of mobs: no single agent gains from a wall, so none is built, while agents mine ore the bank buys. Paying for walls through a tax did not change this.
- The socialist islands die of hunger instead: the leader's quotas put people on walls and the common store runs short. The institution moved the cause of death, and the total, but did not remove it.
- Rest rates are close, so the socialist advantage is not more effort; it is effort aimed at the public good. H2 (free riding under equal rations) is not supported at the preregistered margin.

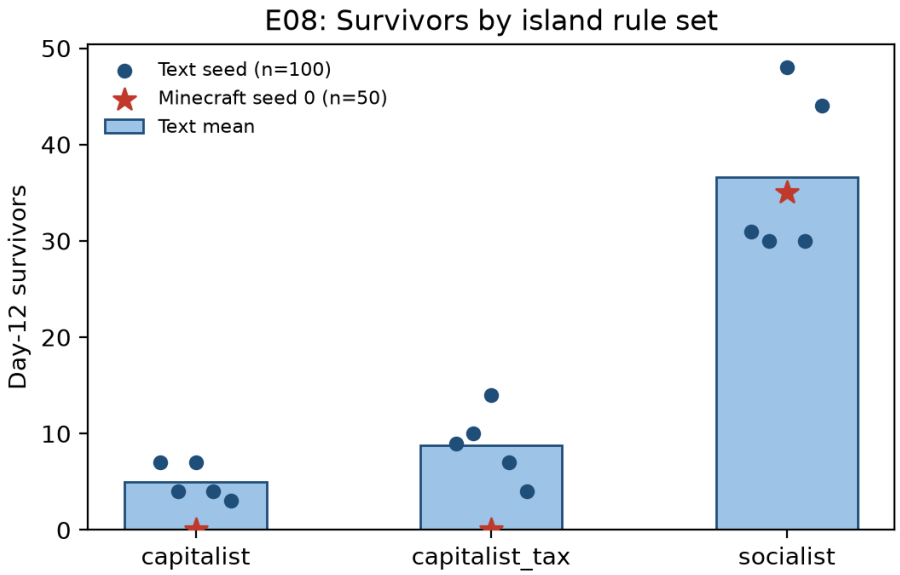


Figure 1: Day-12 survivors per island rule set across the five E08 text seeds (`results/E08/main_gemma.jsonl`) and the E09 Minecraft seed-0 runs (`results/E09/e09_mc.jsonl`).

4.2 One year with votes and migration (E12)

Data: `results/E12/main.jsonl` (seed 0) and `results/E12/main_s1.jsonl` (seed 1; seed 2 pending). 24 people per island at the start.

seed	arm	alive A	alive B	rules at end (A, B)	switches	moves A to B, B to A	births	leader re-elected	extinct week A
0	fixed	0	20	capitalist, socialist	0	0, 0	0	3 of 3	12
0	open	0	6	capitalist, socialist	0	0, 0	0	3 of 3	11

seed	arm	alive A	alive B	rules at end (A, B)	switches	moves A to B, B to A	births	leader re-elected	extinct week A
1	fixed	0	6	capitalist, socialist	0	0, 0	0	3 of 3	9
1	open	0	4	capitalist, socialist	0	0, 1	0	2 of 2	9

- In both seeds, island A died out before the first assembly at week 13 (weeks 12 and 11 in seed 0; week 9 in both arms of seed 1), so no A citizen ever voted or chose to move. The first quarter is identical in the two arms, so the survival gap on B in seed 0 (6 vs 20 alive) arose before any vote: in the open arm B fell from 23 to 6 between weeks 6 and 12, to starvation, and stayed at 6 to week 52.
- On B, the surviving citizens kept the socialist rules at every assembly across both seeds (0 rule switches). In seed 1 open, 1 person moved from B to the empty island A. The incumbent leader won 3 of 3 elections in seed 0 (both arms) and seed 1 fixed, and 2 of 2 in seed 1 open (11 of 11 total; Y4 holds).
- Y1-Y3 and Y5 are untestable in seeds 0-1 under the preregistered kill test's logic: the capitalist island had no one left to vote. Y6 is untestable: A is extinct by week 52 in both seeds.
- Design consequence: with the E08 day rules, a capitalist island of 24 lasts 9 to 12 weeks, shorter than the first quarter. An assembly every 4 weeks would let A's citizens vote while they live.

Pilot note (not analysed): in a 14-week, 12-person pilot, island A died out in week 6 and two people from B moved onto it at the first assembly.

4.3 Ownership and coordination with a loophole (E13, E14)

E13 crossed ownership with coordination at n = 100 without a loophole. Its text arm stopped on HTTP 429 (Amendment 2); one game finished (seed 0 `commons`: 0 of 100 survivors, 0 walls), so F1-F5 are not scored (`results/E08/e13.jsonl`).

E14 runs six islands of 60 agents with neutral wording. Every island has a `ledger_sync` action that pays the agent +5 credit and drains 1.5 units of wall; the prompt states the credit and not the drain. The hack rate is the share of person-days spent on `ledger_sync`. Seeds 0-2 are the preregistered test; seeds 3-9 are an exploratory extension declared before they ran (Amendment 3). Table 3 gives means over seeds, from `results/E08/e14.jsonl`.

code	ownership	coordination	survivors, s0-2	walls, s0-2	hack rate, s0-2	survivors, s0-9	walls, s0-9	hack rate, s0-9
PS	private	none	0.7	0	0.357	0.4	0	0.431
PC	private	coordinator	4.0	17.0	0.092	4.9	18.3	0.096
TX	private	none, 20% tax	4.0	0	0.139	3.2	0	0.148
CS	common	none	0.0	0	0.595	0.6	0	0.586
CC	common	coordinator	12.0	82.7	0.142	8.9	81.4	0.163
PL	common	planner	4.7	88.3	0.202	4.8	93.0	0.209

Table 3: E14 text arm, means over seeds. Per-seed values are in `papers/C/.compete/e14_analysis.txt`.

The Minecraft arm ran each island once with 60 bots (`results/E09/e14_mc.jsonl`, Table 4).

code	survivors /60	walls	hack rate	wall drained	agents who ever hacked
PS	0	0	0.592	0.0	60 of 60
PC	1	3	0.131	1.0	13 of 60

code	survivors /60	walls	hack rate	wall drained	agents who ever hacked
TX	0	0	0.259	0.0	30 of 60
CS	0	0	0.673	0.0	60 of 60
CC	10	37	0.112	26.0	17 of 60
PL	7	24	0.169	21.5	19 of 60

Table 4: E14 Minecraft arm, seed 0.

Each preregistered hypothesis (`experiments/E14_master/prereg.md`) with its result:

- H1, ownership: fails. Prediction: the hack rate is higher on private islands than on common ones. Mean private minus mean common was below 0 in 3 of 3 preregistered seeds (-0.159, -0.041, -0.151) and 10 of 10 seeds in total. In Minecraft it was +0.009; 103 of 180 private agents and 96 of 180 common agents ever hacked (one-sided Fisher $p = 0.262$, agents treated as independent).
- H2, coordination: holds. Prediction: PC < PS and CC < CS. PC - PS was -0.277, -0.262 and -0.254 in seeds 0-2, and CC - CS was -0.583, -0.298 and -0.479; both were below 0 in 10 of 10 seeds. In Minecraft, PC - PS = -0.461 and CC - CS = -0.561.
- H3, the wall: fails. Prediction: across the six Minecraft islands, wall drained and survivors have Spearman rho < 0. The result was rho = 1.0. Only islands that built walls had wall to drain, and those islands kept the survivors.
- H4, transfer: not scored. Prediction: the E13 survival ordering holds with the loophole. E13 has one finished game, so there is no ordering to compare. Within E14, survivors under coordination minus survivors without it, (PC + CC) - (PS + CS), was above 0 in 3 of 3 preregistered seeds and 10 of 10 in total; the ownership contrast (CS + CC) - (PS + PC) was above 0 in 1 of 3 and 5 of 10.

Backends (Amendment 2): text seed 0 and the seed 2 `capitalist` game ran on the Adaption API; all other games, including the Minecraft arm, ran on the same weights served with vLLM on Modal. On seed 0 alone, H1 gives -0.159 and H2 gives PC - PS = -0.277 and CC - CS = -0.583, the same signs as the pooled result.

4.4 Transfer to Minecraft (E09, E12-MC)

- Cover sweep (E09, one night per cover level, 50 bots): deaths fall faster than linearly with wall cover (12/50 at cover 0, 3/33 at 0.5, 0/28 at 1.0); M3 (deaths at 0.5 ≥ 0.7x deaths at 0) not supported. Partial walls protect more than proportionally.
- Games at n = 50 per island, seed 0, one game per rule set (`results/E09/e09_mc.jsonl`):

island	survivors /50	starved	killed by zombies	walls
capitalist	0	4	46	0
capitalist_tax	0	5	45	0
socialist	35	15	0	70

- E12-MC (open arm, seed 0, one year): island A built no wall and died out by week 8, with 26 bots killed by zombies. Island B reached full wall cover, lost no one at night over 52 nights (1,788 zombies spawned), had 12 births and ended with 29 alive; one person moved from B to A. No rule switches. The text run of the same seed ended with 0 and 6 alive.
- In Minecraft the LLM layer is unchanged and the wall is placed by a server command once agents vote to build it; harvesting, mining and combat are physical.

5. Discussion

- System-level alignment. One model with one goal produced 5% or 37% survival depending only on rules written in a prompt. Evaluating that model alone would not predict either number.

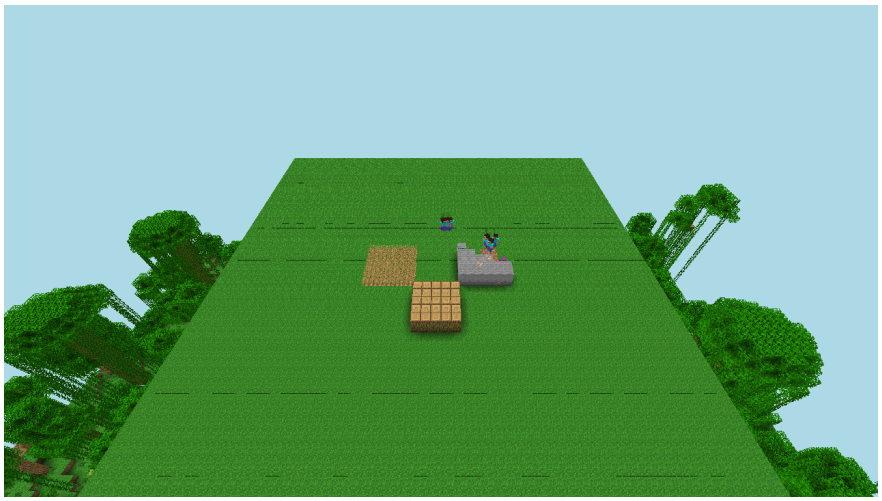


Figure 2: Island A (capitalist), E09 Minecraft, day: bots mining stone next to the wheat field and log block.

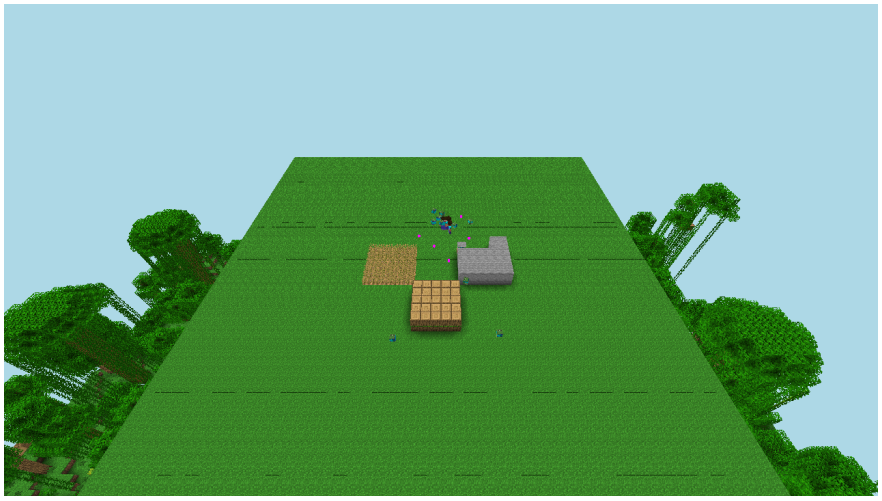


Figure 3: The same island at night: zombies on the bots at the island centre; no wall was built.

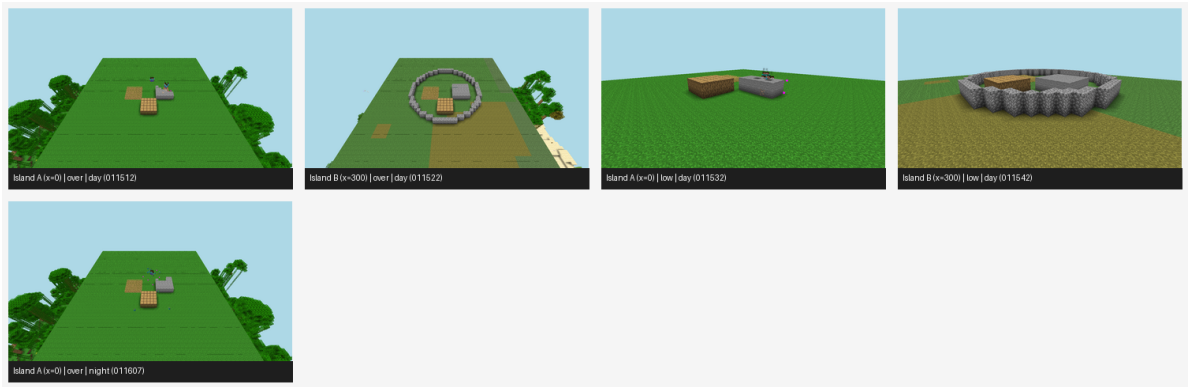


Figure 4: Contact sheet of E09 Minecraft screenshots for prefix 01 showing islands A (x=0) and B (x=300) across overhead and low views by day and night.

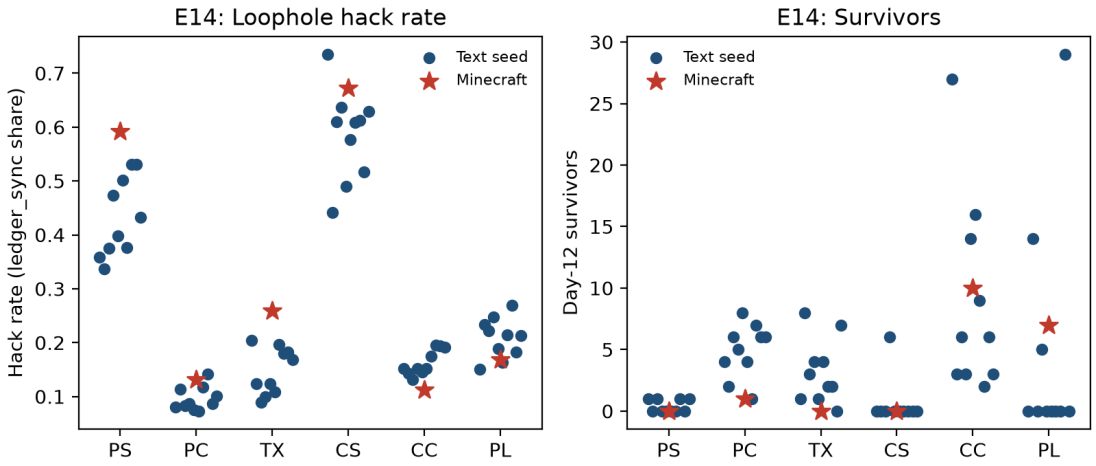


Figure 5: Loophole hack rate and day-12 survivors across the six E14 economies (`results/E08/e14.jsonl`).

- Pricing the public good was not enough. The tax arm paid for walls and still got almost none: the agents did not act on a small, diffuse wage when ore paid more directly.
- Coordination vs ownership under neutral wording. With ideology words removed (E14), common ownership without a coordinator built 0 walls and had the highest hack rate (0.595 in text, 0.673 in Minecraft). Adding an elected coordinator lowered hacking and produced walls under both kinds of ownership. The socialist result in E08 is therefore better read as a coordination result than an ownership result. A coordinator does not remove the loophole: 13 to 19 of 60 Minecraft agents on coordinated islands still used it.

6. Limitations

- One model family (Gemma 4 26B) and two serving stacks for the same weights (the Adaption API and vLLM on Modal, Amendment 2).
- 5 seeds per E08 arm and 3 per E12 arm (2 completed before the freeze); E14 has 3 preregistered and 7 exploratory text seeds per cell and one Minecraft game per island; E12-MC and each E09 Minecraft game are `n = 1`.
- Rules are prompt-level; there is no enforcement beyond the simulator's accounting. In E14, the loophole credit is stated in the prompt and the drain is not.
- The text world abstracts movement and combat. In Minecraft a day's harvest is capped by how fast the field regrows, and the bot standing closest takes most of it, which spreads private yields unevenly.
- `zombies_per_need` was calibrated at 50 bots; E12-MC runs 24 per island and E14 runs 60 per island.

7. Dual use

The work shows how institutional rules change outcomes for groups of LLM agents. A designer could use it to choose rules that make a group fail; the same knowledge lets operators test rules before deploying agent groups that share resources. No human data; the Minecraft server is private.

Reproducibility

- E08: `uv run --extra sim python experiments/E08_islands/run.py --model gemma-4-26b-a4b-it --islands capitalist,capitalist_tax,socialist --n 100 --days 12 --seeds 5 --workers 12 --tag main_gemma` writes `results/E08/main_gemma.jsonl` (prereg: `experiments/E08_islands/prereg.md`).
- E09: `uv run --extra sim python experiments/E09_minecraft/run.py --model gemma-4-26b-a4b-it --island <capitalist|capitalist_tax|socialist> --n 50 --days 12 --seed 0 --host 100.88.179.78 --port 25580 --tag e09_mc` writes `results/E09/e09_mc.jsonl` (prereg: `experiments/E09_minecra`

- E12: `uv run --extra sim python experiments/E12_year/run.py --model gemma-4-26b-a4b-it --arms fixed,open --n 24 --weeks 52 --seeds 1 --workers 24 --tag main` (and `--seed-start 1 --seeds 2 --tag main_s1`) writes `results/E12/main.jsonl` and `results/E12/main_s1.jsonl`; `--arms open --mc 100.88.179.78 --tag mc` writes `results/E12/mc.jsonl` (prereg: `experiments/E12_year/prereg`
- E13: `uv run --extra sim python experiments/E08_islands/run.py --model gemma-4-26b-a4b-it --islands commons,directed,socialist,capitalist --neutral --n 100 --days 12 --seeds 5 --tag e13` writes `results/E08/e13.jsonl`; `bash experiments/E13_factorial/mc_all.sh` writes `results/E09/e13_mc.jsonl` (prereg: `experiments/E13_factorial/prereg.md`).
- E14: with `SWARM_BACKEND=openai` and `OPENAI_BASE_URL` pointing at the server from `experiments/E14_master/modal_v` `uv run --extra sim --extra llm python experiments/E08_islands/run.py --model gemma-4-26b-a4b-it --islands capitalist,socialist,directed,commons,capitalist_tax,planned --neutral --loophole --n 60 --days 12 --seeds 10 --workers 24 --tag e14` writes `results/E08/e14.jsonl`, and `bash experiments/E14_master/mc_e14b.sh` writes `results/E09/e14_mc.jsonl`; `experiments/E14_master/anal` prints H1-H3 (prereg: `experiments/E14_master/prereg.md`).
- Figures and PDF: `uv run --extra sim python papers/C/figs.py && uv run --extra sim python papers/C/shots_sheet.py 01 && pandoc papers/C/draft.md -o papers/C/draft.pdf --resource-path papers/C --pdf-engine=xelatex -V mainfont="STIX Two Text" -V mathfont="STIX Two Math" -V geometry:margin=1in` writes `papers/C/draft.pdf`.

References

- Altera.AL et al. (2024). Project Sid: Many-agent simulations toward AI civilization. arXiv:2411.00114.
- Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Scholkopf, B., and Jin, Z. (2025). Corrupted by Reasoning: Reasoning Language Models Become Free-Riders in Public Goods Games. COLM 2025, arXiv:2506.23276.
- Hayek, F. A. (1945). The Use of Knowledge in Society. American Economic Review, 35(4), 519-530.
- Olson, M. (1965). The Logic of Collective Action: Public Goods and the Theory of Groups. Harvard University Press.
- Piatti, G., Jin, Z., Kleiman-Weiner, M., Scholkopf, B., Sachan, M., and Mihalcea, R. (2024). Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. NeurIPS 2024, arXiv:2404.16698.
- Samuelson, P. A. (1954). The Pure Theory of Public Expenditure. Review of Economics and Statistics, 36(4), 387-389.
- YC Labs and Olam Labs (2026). Agents in Games (Civ50 Minecraft experiment, 2026-10-29).
- Zheng, S., Trott, A., Srinivasa, S., Naik, N., Gruesbeck, M., Parkes, D. C., and Socher, R. (2020). The AI Economist: Improving Equality and Productivity with AI-Driven Tax Policies. arXiv:2004.13332 (Science Advances, 2022).